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From a multi-period perspective: A periodic dynamics forecasting network for multivariate time series forecasting

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ABSTRACT

To achieve more accurate multivariate time series (MTS) forecasting, it is crucial to extract temporal features of individual univariate series and interdependencies among multiple variables that evolve over time (i.e., dynamic interdependencies). Many MTS manifest inherent periodic fluctuations, which are highly significant for effective modeling. However, existing methods for modeling periodic features mainly focus on multi-period temporal features of univariate series and have not sufficiently addressed complex interdependencies in multiple periods, including dynamic dependencies among variables within the same period (Intra-period) and across different periods (Inter-period). Furthermore, methods for modeling dynamic interdependencies are insufficient for mining the inherent multi-periodicity of MTS. Thus, collaboratively capturing both intra-period and inter-period interdependencies remains challenging. To address above challenges, this paper introduces the Periodic Dynamics Forecasting Network (**PDFNet**) for modeling multi-period dynamic interdependencies of MTS. We design a periodic feature extraction module that utilizes frequency domain analysis to identify the multi-period features of MTS. The multi-period temporal networks module is designed to capture temporal features within and across periods from a multi-period perspective. To capture intra-period and inter-period dynamic dependencies among multiple variables, we propose a gated periodic recurrent unit and a gated graph structure learning module to construct dynamic graphs and then effectively learn intra-period and inter-period information through dynamic graph convolution networks. Extensive experiments on multiple MTS datasets have demonstrated our superior performance compared with state-of-the-art methods.

1. Introduction

Multivariate time series (MTS) forecasting plays a critical role in addressing the growing demands of the real world, particularly in fields such as energy management [1], traffic flow analysis [2], and weather forecasting [3], providing support for more precise decision-making and resource allocation. MTS forecasting tasks encounter two significant aspects: temporal features and interdependencies [4]. Temporal features involve learning complex temporal patterns, including periodicity and trend features [5]. Meanwhile, interdependencies require modeling dependencies among multiple variables [6]. Despite significant progress in MTS forecasting, accurately modeling its learning paradigms remains a major challenge due to the inherent complexity of these patterns.

Common methods for learning temporal features include those based on Recurrent Neural Networks (RNNs, e.g., DeepAR [7]), Convolutional Neural Networks (CNNs, e.g., TCN [8] and TimesNet [9]), which excel at capturing local information of univariate series, and Transformers (e.g., FEDformer [10] and Crossformer [11]) that exhibit strong capabilities in capturing temporal long-term features. Current methods considering multi-period or multi-scale characteristics (e.g., TimesNet and PathFormer [12]) predominantly emphasize the extraction and modeling of temporal features; however, their modeling of interdependencies among variables remains implicit.

To comprehensively capture the dependencies in MTS, it is essential to learn not only temporal features but also interdependencies. Among the various methods for capturing interdependencies, graph neural network-based (GNNs, e.g., SteamGNN [13] and FourierGNN [14]) models have achieved competitive performance. These methods treat each time series as a node and the interdependencies as edges, thereby constructing a graph. Due to their strong capability to explicitly capture nonlinear interdependencies, GNNs have significantly enhanced the performance of MTS forecasting. Previous models (e.g., DCRNN [15] and STGNN [16]) heavily rely on predefined graph learning structures, requiring prior knowledge or fixed rules for their construction, which

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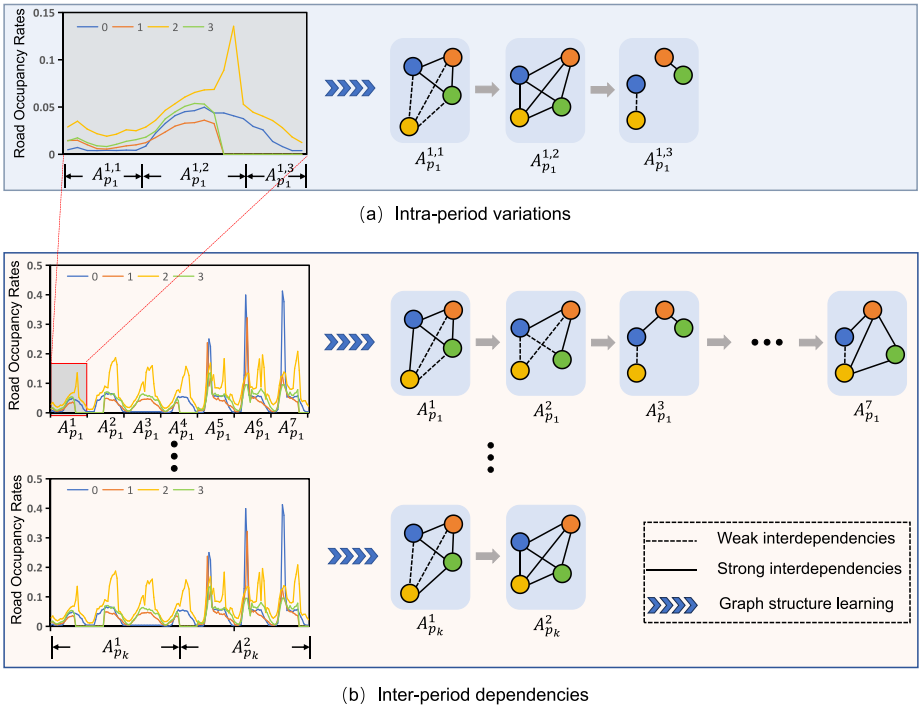


Fig. 1. In the traffic dataset, the changes in road occupancy rates for sensors 0, 1, 2, and 3 from 2:00 AM on July 1, 2016, to 2:00 AM on July 8, 2016. Solid lines represent strong interdependencies, while dashed lines indicate weak interdependencies. $A_{p_k}^{j,l}$ represents the adjacency matrix, where the subscript p_k denotes the period, j indicates the jth period, and l represents the lth stage within the period.

makes it difficult to adapt flexibly to MTS tasks across various domains. Subsequent research (e.g., MTGNN [17] and SteamGNN [18]) has advanced MTS forecasting by adaptively learning adjacency matrices in a data-driven manner, but these studies have not taken into account dynamic dependencies among multiple variables, making it difficult to accurately model interdependencies of MTS in real-world scenarios. Existing methods (e.g., ESG [19], SDGL [20], and DLF [21]) typically focus on learning the evolution of interdependencies but are weak in accounting for crucial multi-period features inherent in MTS.

The multi-periodicity in MTS is embedded in both temporal features and interdependencies among variables. Therefore, extracting the multi-periodicity in these two aspects is crucial. By effectively modeling multi-periodicity, it is possible to better capture the dynamic features of each variable over time, as well as reveal the interdependencies among variables that evolve across different periods. Therefore, mining dynamic intra-period variations and inter-dependencies are important, but the following challenges still exist:

I **Intra-period variations**. Even within the same period, the interdependencies among multiple variables change across different stages. For example, the traffic network structure can differ between morning rush hour and off-peak periods, even on the same working day. Fig. 1(a) shows the daily road occupancy rates recorded by four sensors. In stage $A_{p_1}^{1,1}$, four sensors change in the same trend, and the learned graph structure indicates that there are interdependencies among variables. Solid lines represent strong interdependencies, while dashed lines indicate weak interdependencies. In stage $A_{p_1}^{1,2}$, the four sensors influence each other, initially rising then declining, which strengthens the interdependencies among variables. In the final stage, the four sensors approach zero at different rates, significantly weakening interdependencies among the variables. This indicates that dynamic dependencies continue to occur within a timeframe considered a single period, necessitating fine-grained modeling. Thus *challenge I* is: *how to capture intra-period dependencies among multiple variables*?

II **Inter-period interdependencies**. The interdependencies among multiple variables change across different periods. For example, in transportation, road occupancy at different locations during weekdays is closely linked due to commuting. On weekends, leisure activities change these patterns, altering the dependencies between locations. As shown in Fig. 1(b), periods $A_{p_1}^1$ and $A_{p_k}^7$ have different graph structures. Therefore, simply assuming that the interdependencies among variables remain constant across all periods may prevent the model from capturing actual dynamic interdependencies. Thus *challenge II* is: *how to capture inter-period variations among multiple variables*?

III **Collaborative aggregation**. The intra-period and inter-period interdependencies are constantly changing and influencing each other. For example, As shown in Fig. 1(b), sensors recording road occupancy rates show distinct differences between weekday peak commuting hours ($A_{p_1}^k$) and weekend leisure activity peaks ($A_{p_k}^2$). These variations reflect intra-period and inter-period changes in traffic flow across different times. Thus *challenge III* is: *how to collaborative aggregate intra-period and inter-period information*?

To address the aforementioned challenges, this paper introduces a novel Periodic Dynamics Forecasting Network (**PDFNet**) specifically designed to collaboratively and adaptively capture both intra-period variations and inter-period dynamic dependencies. Unlike previous methods, PDFNet explicitly captures interdependencies between variables across multiple periods through the combination of frequency-domain analysis and dynamic graph modeling techniques. Specifically, we first design a periodic feature extraction (PFE) module to adaptively select the most prominent frequency-domain periodicities based on their energy contributions, facilitating subsequent modeling. Next, we propose a multi-period temporal network consisting of two distinct modules: one employs convolutional operations guided by PFE to capture intra-period temporal features within individual stages; the other utilizes self-attention mechanisms, also guided by PFE, to capture inter-period temporal dependencies across different periods. Additionally,

we introduce a gated periodic recurrent unit to encode hidden representations that reflect the dynamic evolution of these dependencies. Finally, we design a gated graph structure learning module leveraging these hidden representations to dynamically construct graphs, thus effectively aggregating intra-period and inter-period information through dynamic graph convolution networks. Our contributions can be summarized as:

Nomenclature

X	Multivariate time series data matrix of size $N \times T$	N	Number of variables in the time series
T	Number of historical time steps considered	H	Number of future time steps to predict
t	A specific time step in the time series	V	Set of nodes, each corresponding to a variable
E	Set of edges representing relationships between variables	A	General adjacency matrix representing variable relationships
G	Graph representing interdependencies among variables	p_i	Period length of the ith partition
$G^j_{p_i}$	Graph representing interdependencies during the jth period of partition p_i	$A^j_{p_i}$	Adjacency matrix for the jth period in partition p_i
$G^{j,l}_{p_i}$	Graph representing interdependencies during the lth stage of the jth period in partition p_i	$A^{j,l}_{p_i}$	Adjacency matrix for the lth stage of the jth period in partition p_i

- A periodic feature extraction module utilizing frequency domain analysis to adaptively identify prominent periodicities.
- A novel gated periodic recurrent unit designed specifically to capture evolving hidden representations across different periods and stages.
- A gated dynamic graph structure learning module that uniquely integrates periodic temporal representations to dynamically construct evolving graphs, enabling robust modeling of intra- and inter-period dependencies.
- Extensive experiments on multiple multivariate time series datasets demonstrate that our method exhibits significant effectiveness compared to state-of-the-art models.

2. Related work

2.1. Time series forecasting

Early traditional time series forecasting primarily relied on statistical methods. For example, the Autoregressive (AR) [22] model assumes that the current value of a time series is a linear combination of its past values. By fitting linear relationships from historical data, the model predicts future values based on past observations. The Vector Autoregression (VAR) [23] model extends the AR model to multivariate time series forecasting. In recent years, deep learning methods for time series forecasting have developed rapidly, demonstrating powerful nonlinear modeling capabilities, particularly in handling complex, high-dimensional time series problems [24]. Recurrent Neural Networks (RNNs) effectively process sequences with long-time steps through their recurrent structures [7]. Convolutional Neural Networks (CNNs) extract local patterns and features within time series through convolution operations [25]. The Transformer model, utilizing self-attention mechanisms, processes the entire time series in parallel, enabling more effective capture of long-range dependencies [11]. Recent studies have begun to emphasize periodic or multi-scale features as fundamental patterns in time series analysis. For instance, DEPTS [26] instantiates the periodic function as a series of cosine functions to capture diverse periodicities. TimesNet [9] employs Fourier transforms to segment time series into multiple periodic scales for feature extraction. PathFormer leverages dual attention mechanisms to explicitly model intra-period and inter-period temporal dependencies within multivariate time series. However, these models primarily concentrate on modeling temporal features across multiple periods or scales, leaving the interdependencies among variables implicitly represented and thus not explicitly captured.

2.2. Graph neural networks for multivariate time series forecasting

Multivariate time series forecasting involves predicting future values of multiple interrelated variables, which may have complex interdependencies [27]. Graph Neural Networks (GNNs) can efficiently model inter-series relationships, especially when there are spatial, network, or other structural connections, significantly enhancing predictive performance [28]. Previously, many GNNs constructed predefined graphs based on prior knowledge, typically built manually or through domain expertise, thus heavily relying on such prior information [15]. Consequently, some methods have attempted to construct static graphs by mining relationships between variables from data [29]. However, static graphs are limited in that they assume relationships between variables remain constant throughout the entire time series, making it difficult to adapt to dependencies that dynamically change over time [30]. Recent studies [31] learn dynamic graphs to capture the dynamic changes within time series, updating node and edge information with each time step. ESG [19] tackles this issue by dividing historical time series into subsequences based on predetermined scales and employing these subsequences to ascertain the corresponding graph structures. Nevertheless, these methods learn the evolution of graph structures and make it difficult to fully consider the multi-period features in time series—the changes within graph structures during each period and interactions between periods.

3. Problem definition

3.1. Preliminary

Fourier transform. The Discrete Fourier Transform (DFT) is a tool that converts discrete time-domain signals into their frequency-domain representation, defined as:

$$X[k] = \sum_{t=0}^{T-1} x[t] \cdot e^{-i2\pi kt/T}, \quad k = 0, 1, 2, \ldots, T-1, \tag{1}$$

where $x[t]$ is a signal sequence of length T. Computing the DFT directly requires $O(T^2)$ computational complexity, which is costly for large-scale data processing. The Fast Fourier Transform (FFT) is an efficient algorithm for calculating the DFT, leveraging symmetries and periodicities to reduce the complexity to $O(T \log T)$, thereby significantly enhancing computational efficiency.

For real-valued signals, the frequency spectrum exhibits conjugate symmetry, i.e., $X[T-k] = X[k]^*$, where * denotes complex conjugation. This means the second half of the frequency-domain data is a mirror image of the first half. In practical applications, the effective frequency components after the transform only need to consider positions from $k = 0$ to $k = \frac{T}{2}$ (when T is even), totaling $\frac{T}{2} + 1$ points.

3.2. Definition

Graph. We define a graph $G = (V, E, A)$ to represent the interdependencies among multivariate time series, where V is the set of all nodes, each corresponding to a variable, and E is the set of all edges. The adjacency matrix A is an $N \times N$ matrix, where N is the number of variables. The elements of this matrix, $A[i][j]$, quantify the strength of the relationship from variable i to j, which may be expressed as correlation coefficients or measures of causal strength.

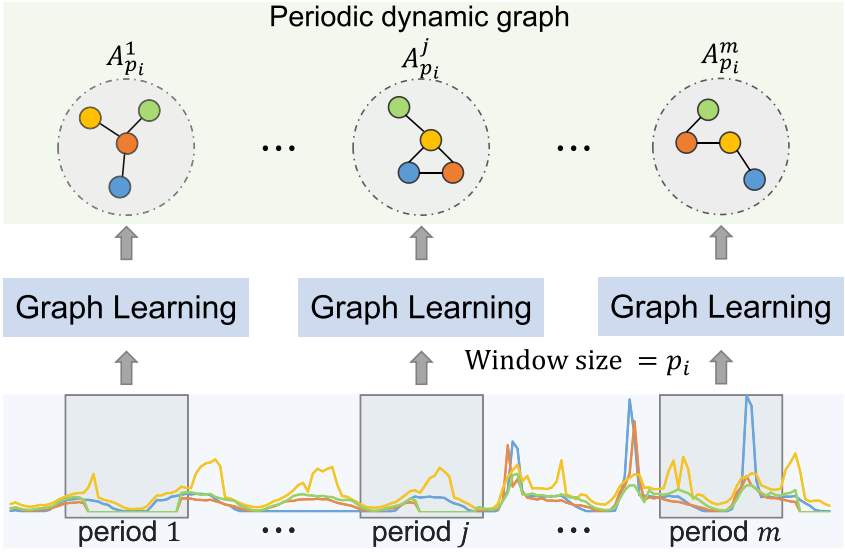


Fig. 2. Periodic dynamic graph.

Periodic dynamic graph. As illustrated in Fig. 2, we define the periodic dynamic graph structures to capture the dynamics of multivariate time series across different periods and stages. For each period, we define the graph structure $G^j_{p_i} = (V, E, A^j_{p_i})$, where $j = 1, 2, \ldots, m$, and $A^j_{p_i}$ is an adjacency matrix that represents the interdependencies among variables during the jth period.

3.3. Problem statement

The multivariate time series forecasting problem is defined as predicting future values of multiple interdependent variables over a specified horizon using their historical observations. In this context, each variable represents a time-dependent feature, and collectively, they form a multivariate time series capturing the dynamics of a system. At each time step t, the system's state is represented by a vector $x_t \in \mathbb{R}^N$, where N is the number of variables. Given the historical data $X = [x_{t-T+1}, x_{t-T+2}, \ldots, x_t] \in \mathbb{R}^{N\times T}$, where T denotes the number of past time steps, the goal is to predict the future values for the next H time steps. This results in a forecasted sequence $\hat{X} = [\hat{x}_{t+1}, \hat{x}_{t+2}, \ldots, \hat{x}_{t+H}] \in \mathbb{R}^{N\times H}$. The task involves constructing a function f that maps the historical multivariate time series data to future predictions:

$$\hat{X}(t+1 : t+H) = f(X(t-T+1 : t)). \tag{2}$$

The objective is to accurately forecast the future states of the multivariate system, capturing both temporal patterns and inter-variable dependencies to facilitate informed decision-making and planning.

4. Methodology

In this section, we elaborate on the comprehensive framework we propose, as illustrated in Fig. 3.

4.1. Periodic feature extraction

Periodicity is a core feature in MTS, reflecting repetitive patterns in the data at fixed time intervals. Moreover, when MTS data exhibit periodicity, the variations in their graph structures often show regular patterns corresponding to these periods. By identifying periodic patterns in MTS, future trends and changes can be predicted more accurately.

Given a multivariate time series (MTS) input $X \in \mathbb{R}^{N\times T}$, we use a Multi-Layer Perceptron (MLP) to embed each variable individually, generating a new embedded representation $X_{\text{embed}} \in \mathbb{R}^{N\times D}$. This embedding facilitates the learning of intra-period and inter-period temporal features. We first need to identify multiple periods in the MTS. Inspired by TimesNet [9], we apply the Real Fast Fourier Transform (rFFT) independently to each variable in X_{embed} and calculate the spectrum (positive frequency part) of the MTS by averaging the results among all variables. The formula for calculating amplitude is as follows:

$$S = \text{Avg}(\text{Abs}(\text{rFFT}(X_{\text{embed}}))). \tag{3}$$

Here, the Abs function calculates amplitude values, and the Avg function computes averages. Together, these functions are used to determine the amplitude value $S \in \mathbb{R}^{D/2}$ for each frequency.

After obtaining the amplitude vector S, we identify the top k frequencies with the highest amplitudes. These frequencies are then used to calculate their corresponding periods, representing the most significant top k periods. The formulas for calculating frequency and period are as follows:

$$f_1, \ldots, f_k = \underset{f_* \in \{1,\ldots,\frac{D}{2}\}}{\text{argTopk}}(S), p_i = \frac{D}{f_i}. \tag{4}$$

To learn inter-period information, based on the periodic length p_i, the time series is divided into m periods:

$$\mathcal{X}^j_{p_i} = \text{Segment}_{p_i}(X_{embed}), j \in \{1, 2, \ldots, m\} \tag{5}$$

where Segment_{p_i} represents the segmentation of periods based on p_i. Next, we also use rFFT to learn the most prominent frequencies and periods within a period. Finally, we divide $\mathcal{X}^l_{p_i}$ into n patches for learning intra-period information. The formulas are as follows:

$$\begin{aligned} S &= \text{Avg}(\text{Abs}(\text{rFFT}(\text{Padding}(\mathcal{X}^j_{p_i})))), \\ p &= \frac{D}{\text{argTop}(S)}, \\ \mathcal{X}^{j,l}_{p_i} &= \text{Segment}_p(\text{Padding}(\mathcal{X}^j_{p_i})), l \in \{1, 2, \ldots, n\}, \end{aligned} \tag{6}$$

where Padding extends the length of the time series by adding zeros, making it suitable for one-dimension convolution. Segment_p represents the segmentation of patches based on p.

Consequently, we obtain representations for each period $\mathcal{X}^j_{p_i}$ and each patch $\mathcal{X}^{j,l}_{p_i}$, which are used to learn intra-period variations and inter-period interdependencies. For convenience, we omit the subscript p_i in subsequent representations.

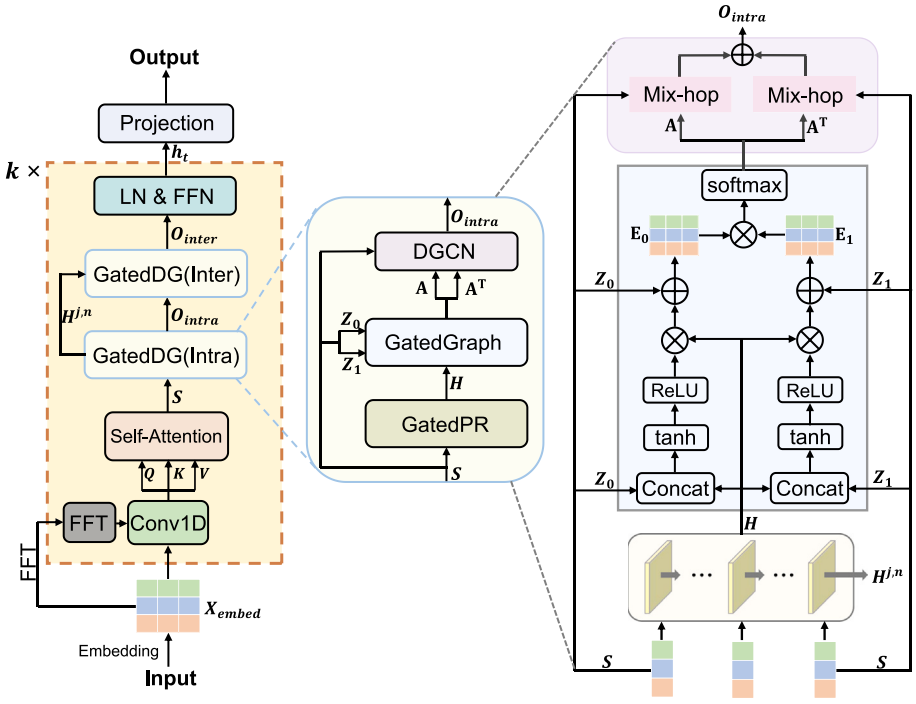


Fig. 3. Overall structure of PDFNet. GatedPR represents the gated periodic recurrent unit. GatedGraph represents the gated graph structure learning module. DGCN stands for dynamic graph convolution network. LN and FFN refer to layer normalization and feed forward networks, respectively.

4.2. Multi-period temporal network

Learning temporal features helps us understand and predict future behaviors of time series more accurately, including both intra-period and inter-period temporal features. First, One-dimensional convolution is employed to learn intra-period temporal features within each period:

$$\mathcal{X}^j = \text{Conv1D}(\text{Padding}(\mathcal{X}^j)), j \in \{1, 2, \dots, m\}. \tag{7}$$

Here, $\mathcal{X}^j \in \mathbb{R}^{N \times D}$ represents intra-period temporal representation of the jth period after convolution.

To learn inter-period temporal representations, we employ a self-attention mechanism to aggregate the m periodic representations and capture long-term temporal features. The formulas are as follows:

$$\begin{aligned} &\mathcal{P} = \text{Truncate}(\text{Concat}(\mathcal{X}^1, \mathcal{X}^2, \dots, \mathcal{X}^m)) \\ &\mathbf{Q} = \mathcal{P}\mathbf{W}_Q, \quad \mathbf{K} = \mathcal{P}\mathbf{W}_K, \quad \mathbf{V} = \mathcal{P}\mathbf{W}_V, \\ &S = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}}\right)\mathbf{V} \end{aligned} \tag{8}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times D}$ represent the learnable embedding matrices. We use Truncate to truncate the padded series, restoring it to its original length D. Thus, this process captures intra-period and inter-period temporal features for subsequent gated graph structure learning.

Algorithm 1: PDFNet

Input: Historical MTS $X \in \mathbb{R}^{N \times T}$, look-back window T, horizon H, variates number N, period number k, other hyperparameters.
Output: $\hat{X}(t+1:t+H)$

$X_{embed} = \text{MLP}(X)$;
$i \leftarrow 1$;
$p_1, \cdots, p_k = \text{FFT}(X_{embed}), \quad i \in \{1, \dots, k\}$;
$\mathcal{X}^j_{p_i} = \text{Segment}_{p_i}(X_{embed}), j \in \{1, 2, \cdots, m\}$;
while $i \le k$ **do**
  $S = \text{Avg}(\text{Abs}(\text{rFFT}(\text{Padding}(\mathcal{X}^j_{p_i}))))$;
  $p = \frac{D}{\text{argTop}(S)}$;
  $\mathcal{X}^{j,l}_{p_i} = \text{Segment}_p(\text{Padding}(\mathcal{X}^j_{p_i})), l \in \{1, 2, \cdots, n\}$;
  $\mathcal{X}^j = \text{Conv1D}(\text{Padding}(\mathcal{X}^j)), j \in \{1, 2, \cdots, m\}$;
  $S = \text{Self-Attention}(\mathcal{X})$;
  $\mathcal{O}_{intra}, \mathbf{H}^j = \text{GatedDG}(S)$;
  $\mathcal{O}_{inter} = \text{GatedDG}(\mathbf{H}^j, \mathcal{O}_{intra})$;
  $h_t = \text{LayerNorm}(\text{FeedForward}(\mathcal{O}_{inter}))$;
end
$\hat{X}(t+1:t+H) = \text{MLP}(h_t)$

4.3. Gated dynamic graph structure learning

To capture intra-period and inter-period dependencies among multiple variables, we proposed a gated periodic recurrent unit and a gated graph structure learning module to construct evolving graph structures.

We first capture intra-period dependencies among multiple variables. The matrix $\mathbf{E} = \text{softmax}(\mathbf{Q}\mathbf{K}^T/\sqrt{D})$, learned through self-attention, actually reflects the static interdependencies across entire MTSs. Consequently, $\mathbf{H}^{j,0} = \mathbf{E}\mathbf{V}$ can serve as the global static hidden state fed into the gated periodic recurrent unit. Subsequently, each patch $\mathcal{X}^l$ is input into the gated periodic recurrent unit, producing n dynamic hidden states that represent changes in graph structure at different patches:

$$\begin{aligned} \mathbf{r}^{j,l} &= \sigma(\text{Padding}([S^{j,l}, \mathbf{H}^{j,l-1}])\mathbf{W}_\mathbf{r} + \mathbf{b}_\mathbf{r}), \\ \mathbf{u}^{j,l} &= \sigma(\text{Padding}([S^{j,l}, \mathbf{H}^{j,l-1}])\mathbf{W}_\mathbf{u} + \mathbf{b}_\mathbf{u}), \\ \mathbf{o}^{j,l} &= \mu([S^{j,l}, (\mathbf{r}^{j,l} \odot \mathbf{H}^{j,l-1})]\mathbf{W}_\mathbf{o} + \mathbf{b}_\mathbf{o}), \\ \mathbf{H}^{j,l} &= \mathbf{u}^{j,l} \odot \mathbf{H}^{j,l-1} + (1 - \mathbf{u}^{j,l}) \odot \mathbf{o}^{j,l}, \end{aligned} \tag{9}$$

where $\mathbf{W}_r, \mathbf{W}_u, \mathbf{W}_o \in \mathbb{R}^{2D \times D}$ are learnable embedding matrices. The Padding extends the length of $[S^{j,l}, \mathbf{H}^{j,l-1}]$ to $2D$. $\mathbf{b}_\mathbf{r}, \mathbf{b}_\mathbf{u}, \mathbf{b}_\mathbf{o} \in \mathbb{R}^{N \times D}$ are bias. $\mathbf{r}^{j,l}$ adjusts the amount of past information to forget. $\mathbf{u}^{j,l}$ balances the retention of the previous hidden state $\mathbf{H}^{j,l-1}$ and the incorporation of the candidate hidden state $\mathbf{o}^{j,l}$. $\mathbf{o}^{j,l}$ represents the new information

to be potentially added to the hidden state. σ represents the sigmoid function, μ is the hyperbolic tangent function, and $\odot$ denotes the Hadamard product.

Therefore, we obtain the hidden state $\mathbf{H}^{j,l}$ for each patch, which will be used to construct the graph structure on this patch. We designed a gated graph structure learning module that integrates local and global temporal features to infer variations of the graph structure. The formulas are as follows:

$$\begin{aligned}
&\mathbf{E}_{\alpha}^{j,l} = \mathbf{Z}_{\alpha} + \mathcal{T}_{\alpha}^{j,l} * \mathbf{H}^{j,l},\\
&\mathcal{T}_{\alpha}^{j,l} = \mathrm{ReLU}(\tanh(\mathrm{Padding}([\mathbf{Z}_{\alpha} \parallel \mathbf{H}^{j,l}])\mathbf{W}_{\alpha})),\\
&\mathbf{Z}_{\alpha} = \begin{cases} \mathbf{AV} & \text{if} \quad \alpha = 0\\ \mathbf{A}^{\top}\mathbf{V} & \text{if} \quad \alpha = 1 \end{cases},
\end{aligned} \tag{10}$$

where $W_{\alpha} \in \mathbb{R}^{2D\times 1}$ are learnable parameters. In this way, we obtain the dynamic graph structure for each patch within the current period, explicitly capturing the dynamic interdependencies underlying MTS:

$$\mathbf{A}^{j,l} = \mathrm{Softmax}\left(\mathrm{ReLU}\left(\mathbf{E}_0^{j,l}\mathbf{E}_1^{j,l\top}\right)\right). \tag{11}$$

Finally, based on the learned graph structures, we employ dynamic graph convolution networks to model the dynamic interdependencies within a period of MTS. DGCN consists of two steps: the information propagation step and the information selection step. The information propagation step is defined as follows:

$$\mathcal{S}_{\ell}^{j,l} = \beta S^{j,l} + (1-\beta)\mathbf{A}^{l}\mathcal{S}_{\ell-1}^{j,l}, \tag{12}$$

where β is a hyperparameter that controls the proportion of the MTS's original state that is retained. The information selection step is defined as follows:

$$\begin{aligned}
&\mathcal{S}_{out}^{j,l} = \sum_{\ell=0}^{L} \mathcal{S}_{\ell}^{j,l}\mathbf{W}_{\ell},\\
&\mathcal{O}_{intra} = \mathrm{Concat}(\mathcal{S}_{out}^{1,1}, \mathcal{S}_{out}^{1,2}, \ldots, \mathcal{S}_{out}^{j,l}, \ldots, \mathcal{S}_{out}^{m,n}).
\end{aligned} \tag{13}$$

We refer to the aforementioned gated periodic recurrent unit, gated graph structure learning module, and DGCN module collectively as GatedDG. To capture inter-period dependencies, we utilize the last hidden state $\mathbf{H}^{j,n}$ output from each patch to represent the entire period's hidden state $\mathbf{H}^{j}$. Additionally, we use $\mathcal{S}^{j}$ as the input for each period. The formulas are as follows:

$$\begin{aligned}
&\mathcal{S}_{out}^{j} = \mathrm{GatedDG}(\mathbf{H}^{j}, \mathcal{S}^{j}),\\
&\mathcal{O}_{inter} = \mathrm{Concat}(\mathcal{S}_{out}^{1}, \mathcal{S}_{out}^{2}, \ldots, \mathcal{S}_{out}^{j}, \ldots, \mathcal{S}_{out}^{m}).
\end{aligned} \tag{14}$$

GatedDG captures intra-period variations and inter-period dependencies among multiple variables, constructing distinct graphs for different patches and periods. It adapts flexibly to various complex scenarios encountered in real-world settings. We input the output $\mathcal{O}_{inter}$ from the kth layer into layer normalization and feed-forward networks to obtain h_t, which is used for downstream prediction tasks. The pseudocode for our proposed PDFNet is delineated in Algorithm 1. This algorithmic representation aids in detailing the structured approach utilized in handling data processing and model training phases.

4.4. Prediction layer and training

In the final layer of our model, we predict the future values of the multivariate time series by leveraging the learned representations from historical data. Specifically, we apply a Multi-Layer Perceptron (MLP) to the hidden state $\mathbf{h}_t$ obtained at the last historical time step t to generate predictions for the next H time steps:

$$\hat{X}_{t+1:t+H} = \mathrm{MLP}(\mathbf{h}_t), \tag{15}$$

where $\hat{X}_{t+1:t+H} \in \mathbb{R}^{N\times H}$ represents the predicted values for all N variables over the future H time steps.

To train the model, we adopt the Mean Squared Error (MSE) loss function, which is suitable for continuous value prediction tasks. The MSE loss is defined as:

$$\mathcal{L} = \frac{1}{NH}\sum_{n=1}^{N}\sum_{h=1}^{H}\left(X_{n,t+h} - \hat{X}_{n,t+h}\right)^2, \tag{16}$$

where $X_{n,t+h}$ is the ground truth value of variable n at time $t+h$, $\hat{X}_{n,t+h}$ is the predicted value from the model. By minimizing this loss function, we optimize the model parameters to reduce the discrepancy between the predicted and actual future values across all variables and time steps.

4.5. Computational complexity analysis

The complexity of the periodic feature extraction module is $O(N \log N)$, arising mainly from frequency-domain transformations. The gated periodic recurrent unit (GPRU) has a complexity of $O(N D^2)$, where N denotes the input sequence length, and D is the hidden state dimension. Additionally, the gated dynamic graph learning module incurs a complexity of $O(N^2 d)$, attributed to pairwise computations between node embeddings, with d representing the embedding dimension. Considering D and d as constants, the overall computational complexity of our model simplifies to $O(N^2)$.

5. Experiments

In this section, we conduct an empirical evaluation of the PDFNet model on real-world benchmark datasets, comparing its performance with various effective time series forecasting models, including state-of-the-art models based on graph neural networks. First, we introduce the time series forecasting datasets used, evaluation metrics, and baseline models for comparison. Then, we present the main experimental results and provide an in-depth analysis of the model through additional insights and ablation studies.

5.1. Experimental settings

5.1.1. Datasets

As shown in Table 1, we utilize a range of benchmark datasets from the transportation, energy, electricity, and economy sectors to evaluate the performance of MTS forecasting models. "Variables" refers to the number of variables in each dataset, "Samples" denotes the number of samples within the dataset, "Sample rate" describes the frequency at which data is collected, "Look-back window" defines the length of historical data relied upon for making predictions, "Horizon" indicates the temporal scope of the forecasts, "Split" details the division ratios of the data during training, validation, and testing phases, "Domain" identifies the application area of the dataset, and "Task" describes the type of problem being addressed or the task being performed using the dataset.

- **ILI.** The ILI dataset contains weekly recorded influenza-like illness patient ratios from the Centers for Disease Control and Prevention (CDC), capturing seasonal patterns and epidemic dynamics.
- **Solar Energy.** The solar energy dataset was published by Lai et al. [32] in 2007 and originates from the National Renewable Energy Laboratory in Alabama. It contains records of solar power production from 137 photovoltaic power stations.
- **Traffic.** The traffic dataset, published by Lai et al. [32], covers the period from 2015 to 2016 and originates from the California Department of Transportation. It specifically focuses on the highways in the San Francisco Bay Area and includes hourly road occupancy measurements collected from 862 sensors.
- **Exchange Rate.** The exchange rate dataset, published by Lai et al. [32], spans from 1990 to 2016 and provides daily exchange rate records for eight countries: Australia, the United Kingdom, Canada, Switzerland, China, Japan, New Zealand, and Singapore.

Table 1
Detailed dataset descriptions.

Dataset	Variables	Samples	Sample rate	Look-back window	Horizon	Split	Domain	Task
ILI	7	966	1 week	36	{24, 36, 48, 60}	6:2:1	Medical	Long Term
Solar Energy	137	52560	10 min	96	{96, 192, 336, 720}		Energy	
Exchange Rate	8	7588	1 day				Economy	
ETTh1, ETTh2	7	17420	1 h				Electricity	
ETTm1	7	69680	15 min					
Electricity	7	26304	1 h					
Traffic	862	17544	1 h					
METR-LA	207	34272	5 min	12	{3, 6, 12}	7:2:1	Transportation	Short Term
PEMS-BAY	325	52116						

Table 2
Categorization of models and methods for addressing temporal features and interdependencies.

Two aspects	Methods	Models
Intra-series variations	Transformer	Autoformer [34], FEDformer [10], PatchTST [35], Pathformer [12]
	Linear	TiDE [36], DLinear [37], TimeMixer [38]
	TCN	MICN [39], TimesNet [9]
Inter-series dependencies	Without graphs	Crossformer [11]
	Pre-defined	DCRNN [15]
	Adaptive	Graph WaveNet [40], AGCRN [2], MTGNN [17]
	Dynamic	ESG [19]

- **Electricity.** The electricity dataset, published by Lai et al. [32], covers the period from 2012 to 2014 and is sourced from the UCI Machine Learning Repository. It includes hourly electricity consumption records from 321 customers.
- **ETT.** The ETT dataset, published by Zhou et al. [33], consists of data from two different counties in China, spanning a two-year period. It includes ETTh1 and ETTh2, which record data hourly, and ETTm1, which captures data every 15 min. Each data point includes the target variable 'oil temperature' and six features related to electrical load.
- **METR-LA.** The METR-LA dataset, published by Li et al. [15], spans from March to June 2012 and is sourced from the Los Angeles County Metropolitan Transportation Authority. It includes average traffic speed measurements collected from 207 sensors located along highways in Los Angeles County.
- **PEMS-BAY.** The PEMS-BAY dataset, published by Bai et al. [2], covers the period from January to May 2017 and originates from the California Department of Transportation. It includes average traffic speed measurements collected from 325 sensors across the Bay Area.

5.1.2. Implementation datails

To facilitate fair and consistent comparison with existing state-of-the-art methods, we adopt widely recognized experimental settings commonly used in recent time series forecasting literature (e.g., PathFormer [12], TimesNet [9], PatchTST [35]). Specifically, we set the lookback window length to 36 for the ILI dataset and 96 for all other datasets. For forecasting horizons, we adopt two groups of settings based on dataset characteristics: forecast horizons of 24, 36, 48, and 60 for the ILI dataset due to its relatively shorter length, and forecast horizons of 96, 192, 336, and 720 for all other datasets, which possess longer sequences. Additionally, for short-term forecasting tasks, we set the lookback window to 12, with forecast horizons of {3, 6, 12}, consistent with established practices in previous studies (e.g.,ESG [19], AGCRN [2], MTGNN [17]).

Our model was developed utilizing PyTorch and trained on an NVIDIA A800-SXM4-80 GB GPU to ensure optimal performance. Training optimization was achieved using the ADAM optimizer with learning rates set at $\{1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3\}$, and the MSE loss function was employed to manage training loss. The training regimen consisted of 10 epochs with a batch size of 32. We specifically targeted the top k significant periodic features for feature selection, selecting k values from $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The architecture of the model was configured with a dimensionality of 128 to adequately capture the complexity of the data.

To prevent overfitting and optimize computational resources, training is terminated prematurely if the loss on the validation set does not decrease for three consecutive epochs. For baseline models, we adhere to hyperparameters specified in original studies that conducted experiments on our datasets, such as the number of layers and dimensions of hidden states. In the absence of prior experiments, we systematically determine appropriate hyperparameters through a grid search methodology on the validation set.

5.1.3. Baselines

As shown in Table 2, we use two sets of baselines for multivariate time series forecasting: one focuses primarily on modeling temporal features, while the other is dedicated to capturing interdependencies.

Methods for modeling temporal features. We selected 7 well-established forecasting models as benchmarks, including:

- **Transformer-based methods.** Autoformer [34] introduces a Decomposition Architecture that can extract more predictable components from complex temporal patterns. FEDFormer [10] presents a model architecture that applies the attention mechanism in the frequency domain to enhance robustness against noise. PatchTST [35] improves the capture of local series information by segmenting time series into multiple subseries-level patches. PathFormer [12] employs attention mechanisms to effectively capture both intra-scale and inter-scale temporal dependencies.
- **MLP-based methods.** DLinear [37] introduces a simple yet effective single-layer linear model that outperforms many Transformer-based models. TiDE [36] designs a time-series dense encoder that effectively handles long-term dependencies in time series. TimeMixer [38] utilizes mixing operations to capture both temporal dynamics and feature interactions.

Table 3
Multivariate long-term time series forecasting results on the Exchange, Traffic, and ILI Energy datasets. The best results are highlighted in bold and the second best are underlined.

Models	Metric	Exchange				Traffic				ILI				1st
		96	192	336	720	96	192	336	720	24	36	48	60	
Ours	MSE	**0.085±0.007**	**0.175±0.009**	**0.330±0.001**	0.820±0.001	**0.393±0.001**	**0.412±0.000**	**0.423±0.000**	**0.459±0.001**	2.292±0.001	2.132±0.001	2.232±0.002	2.491±0.000	7
	MAE	**0.205±0.001**	**0.297±0.000**	**0.416±0.000**	0.685±0.001	**0.268±0.000**	**0.277±0.002**	**0.283±0.001**	**0.301±0.001**	0.930±0.000	0.937±0.001	0.956±0.003	1.053±0.001	7
Pathfromer	MSE	0.090±0.000	0.186±0.000	0.404±0.001	0.853±0.001	0.479±0.001	0.484±0.000	0.503±0.000	0.537±0.001	1.718±0.000	**1.768±0.001**	**1.732±0.000**	**1.819±0.001**	3
	MAE	0.208±0.001	0.307±0.000	0.457±0.000	0.700±0.001	0.283±0.002	0.292±0.000	0.299±0.000	0.322±0.000	**0.840±0.002**	**0.825±0.001**	**0.831±0.001**	**0.870±0.000**	4
TimeMixer	MSE	0.091±0.002	0.180±0.001	0.363±0.001	0.867±0.000	0.462±0.001	0.473±0.000	0.498±0.000	0.506±0.002	5.416±0.000	4.627±0.001	4.348±0.000	4.154±0.001	0
	MAE	0.210±0.001	0.299±0.002	0.438±0.000	0.695±0.001	0.285±0.001	0.296±0.003	0.296±0.002	0.313±0.001	1.785±0.002	1.600±0.001	1.556±0.002	1.444±0.000	0
PatchTST	MSE	0.097±0.002	0.181±0.001	0.343±0.000	0.936±0.001	0.459±0.000	0.469±0.000	0.483±0.002	0.517±0.001	2.159±0.001	2.306±0.001	2.037±0.001	2.014±0.000	0
	MAE	0.215±0.001	0.303±0.002	0.426±0.000	0.726±0.000	0.298±0.001	0.301±0.001	0.307±0.002	0.326±0.001	0.876±0.001	0.901±0.002	0.876±0.000	0.913±0.001	0
FEDformer	MSE	0.166±0.000	0.270±0.000	0.473±0.001	1.166±0.000	0.591±0.002	0.608±0.000	0.619±0.001	0.647±0.001	3.158±0.002	3.199±0.000	2.888±0.000	3.132±0.001	0
	MAE	0.294±0.006	0.381±0.001	0.505±0.003	0.829±0.000	0.372±0.000	0.382±0.001	0.384±0.002	0.395±0.001	1.247±0.000	1.232±0.002	1.157±0.001	1.227±0.001	0
Autoformer	MSE	0.143±0.001	0.279±0.000	0.481±0.000	1.115±0.002	0.673±0.003	0.642±0.002	0.637±0.000	0.669±0.000	3.444±0.004	3.366±0.001	3.063±0.001	2.805±0.00	0
	MAE	0.273±0.001	0.385±0.001	0.520±0.000	0.821±0.001	0.406±0.001	0.408±0.003	0.397±0.000	0.413±0.001	1.293±0.001	1.256±0.000	1.179±0.000	1.130±0.002	0
TimesNet	MSE	0.105±0.002	0.245±0.000	0.363±0.002	0.931±0.001	0.589±0.005	0.617±0.000	0.635±0.001	0.660±0.001	**1.638±0.002**	2.678±0.001	2.547±0.001	1.982±0.004	1
	MAE	0.235±0.006	0.360±0.000	0.439±0.002	0.734±0.005	0.315±0.002	0.324±0.001	0.338±0.000	0.350±0.003	0.866±0.000	0.987±0.005	0.933±0.002	0.898±0.018	0
MICN	MSE	0.093±0.001	0.185±0.000	0.354±0.001	0.872±0.000	0.520±0.000	0.537±0.000	0.549±0.000	0.576±0.001	3.398±0.005	2.710±0.000	2.822±0.001	3.031±0.000	0
	MAE	0.228±0.002	0.332±0.007	0.468±0.001	0.727±0.000	0.313±0.003	0.318±0.000	0.322±0.001	0.334±0.001	1.275±0.011	1.125±0.001	1.156±0.003	1.196±0.006	0
TiDE	MSE	0.094±0.007	0.202±0.001	0.349±0.000	0.860±0.004	0.821±0.001	0.772±0.006	0.782±0.000	0.741±0.002	5.934±0.009	5.407±0.002	4.485±0.000	4.580±0.001	0
	MAE	0.217±0.000	0.324±0.000	0.431±0.001	0.701±0.000	0.498±0.005	0.481±0.009	0.497±0.001	0.462±0.000	1.807±0.004	1.702±0.002	1.523±0.010	1.546±0.001	0
DLinear	MSE	0.098±0.003	0.203±0.000	0.343±0.002	**0.749±0.000**	0.697±0.002	0.647±0.000	0.653±0.000	0.694±0.001	5.061±0.021	4.414±0.031	4.109±0.002	4.234±0.028	1
	MAE	0.233±0.005	0.344±0.000	0.448±0.001	**0.664±0.003**	0.429±0.000	0.407±0.000	0.410±0.001	0.429±0.000	1.710±0.000	1.550±0.002	1.473±0.001	1.482±0.004	1

Table 4
Multivariate long-term time series forecasting results on the ETT datasets. The best results are highlighted in bold and the second best are underlined.

Models	Metric	ETTh1				ETTh2				ETTm1				1st
		96	192	336	720	96	192	336	720	96	192	336	720	
Ours	MSE	**0.308±0.002**	0.449±0.001	0.492±0.000	0.522±0.000	0.300±0.001	0.382±0.000	0.424±0.001	**0.426±0.001**	**0.315±0.001**	0.362±0.000	0.395±0.000	0.456±0.001	3
	MAE	0.410±0.001	0.441±0.002	0.465±0.000	0.504±0.001	**0.350±0.001**	**0.400±0.000**	0.432±0.000	**0.445±0.000**	**0.347±0.001**	**0.370±0.000**	**0.392±0.000**	**0.423±0.001**	7
Pathfromer	MSE	0.388±0.001	0.442±0.000	0.482±0.002	**0.489±0.001**	0.291±0.000	**0.361±0.001**	**0.392±0.001**	0.428±0.000	0.321±0.001	0.369±0.002	0.400±0.001	0.474±0.001	3
	MAE	**0.390±0.000**	**0.419±0.001**	**0.438±0.001**	**0.458±0.000**	0.337±0.001	0.383±0.000	0.411±0.002	0.442±0.001	0.350±0.002	0.376±0.000	0.395±0.001	0.433±0.000	4
TimeMixer	MSE	0.375±0.002	0.429±0.000	0.484±0.001	0.498±0.002	**0.287±0.000**	0.379±0.000	0.430±0.001	0.476±0.000	0.319±0.003	0.364±0.001	0.390±0.002	0.451±0.001	1
	MAE	0.400±0.001	0.421±0.001	0.458±0.001	0.482±0.000	0.341±0.002	0.400±0.001	0.441±0.002	0.470±0.000	0.360±0.001	0.386±0.000	0.404±0.000	0.441±0.001	0
PatchTST	MSE	0.377±0.001	0.426±0.000	0.469±0.000	0.519±0.001	0.309±0.001	0.381±0.009	0.412±0.002	0.436±0.011	0.328±0.001	0.377±0.000	0.409±0.001	0.468±0.001	0
	MAE	0.397±0.000	0.432±0.000	0.457±0.001	0.504±0.001	0.359±0.002	0.407±0.000	**0.429±0.001**	0.456±0.000	0.367±0.002	0.386±0.000	0.406±0.000	0.438±0.001	1
FEDformer	MSE	0.377±0.002	**0.420±0.001**	**0.458±0.001**	0.502±0.000	0.344±0.004	0.442±0.001	0.499±0.001	0.480±0.002	0.366±0.000	0.447±0.001	0.467±0.001	0.507±0.003	2
	MAE	0.418±0.003	0.444±0.000	0.467±0.000	0.503±0.001	0.389±0.001	0.450±0.009	0.490±0.002	0.487±0.003	0.412±0.005	0.441±0.001	0.470±0.001	0.489±0.000	0
Autoformer	MSE	0.499±0.001	0.487±0.004	0.524±0.002	0.543±0.001	0.382±0.003	0.451±0.000	0.490±0.005	0.504±0.001	0.650±0.001	0.612±0.007	0.580±0.001	0.643±0.002	0
	MAE	0.481±0.000	0.476±0.002	0.504±0.001	0.523±0.000	0.416±0.007	0.453±0.002	0.486±0.000	0.503±0.006	0.524±0.003	0.520±0.001	0.510±0.001	0.544±0.000	0
TimesNet	MSE	0.389±0.000	0.439±0.009	0.494±0.000	0.518±0.001	0.337±0.001	0.405±0.000	0.454±0.001	0.434±0.001	0.351±0.000	0.380±0.000	0.446±0.003	0.527±0.001	0
	MAE	0.412±0.001	0.442±0.000	0.471±0.000	0.495±0.003	0.371±0.010	0.415±0.002	0.452±0.008	0.448±0.003	0.384±0.001	0.396±0.001	0.438±0.000	0.474±0.000	0
MICN	MSE	0.394±0.001	0.429±0.000	0.464±0.002	0.578±0.000	0.389±0.000	0.523±0.003	0.581±0.007	0.804±0.001	0.338±0.002	**0.355±0.000**	**0.381±0.001**	**0.434±0.000**	3
	MAE	0.421±0.000	0.442±0.001	0.474±0.005	0.559±0.002	0.425±0.006	0.497±0.013	0.534±0.002	0.645±0.006	0.381±0.002	0.388±0.000	0.409±0.000	0.446±0.001	0
TiDE	MSE	0.658±0.009	0.707±0.001	0.735±0.015	0.667±0.003	0.394±0.013	0.499±0.005	0.562±0.009	0.530±0.018	0.366±0.000	0.401±0.000	0.433±0.003	0.495±0.001	0
	MAE	0.536±0.001	0.556±0.000	0.569±0.001	0.560±0.002	0.406±0.018	0.461±0.007	0.502±0.015	0.496±0.010	0.385±0.000	0.401±0.002	0.422±0.001	0.456±0.000	0
DLinear	MSE	0.396±0.000	0.445±0.000	0.487±0.001	0.513±0.005	0.341±0.025	0.482±0.013	0.593±0.020	0.840±0.026	0.346±0.001	0.382±0.000	0.415±0.001	0.473±0.001	0
	MAE	0.411±0.001	0.440±0.001	0.465±0.002	0.510±0.004	0.395±0.024	0.479±0.010	0.542±0.009	0.661±0.010	0.374±0.002	0.391±0.003	0.415±0.002	0.451±0.001	0

Table 5
Multivariate long-term time series forecasting results on the Electricity, Solar and Weather datasets. The best results are highlighted in bold and the second best are underlined.

Models	Metric	Electricity				Solar Energy				Weather				1st
		96	192	336	720	96	192	336	720	96	192	336	720	
Ours	MSE	**0.149±0.000**	**0.164±0.000**	**0.178±0.001**	**0.227±0.001**	0.209±0.003	0.249±0.001	0.254±0.001	0.257±0.000	0.173±0.001	0.225±0.000	0.280±0.001	0.361±0.001	4
	MAE	**0.241±0.001**	**0.256±0.000**	**0.271±0.001**	0.311±0.001	**0.235±0.001**	**0.260±0.000**	**0.274±0.000**	**0.278±0.001**	0.213±0.005	0.258±0.001	**0.258±0.002**	0.351±0.002	8
Pathfromer	MSE	0.161±0.003	0.175±0.000	0.192±0.001	0.232±0.001	0.215±0.002	0.250±0.003	0.259±0.005	0.270±0.002	**0.151±0.001**	**0.204±0.001**	0.262±0.000	0.342±0.000	2
	MAE	0.254±0.001	0.258±0.000	0.275±0.000	**0.307±0.001**	0.243±0.001	0.288±0.001	0.297±0.004	0.301±0.003	**0.191±0.002**	**0.243±0.000**	0.285±0.001	**0.337±0.000**	4
TimeMixer	MSE	0.156±0.000	0.170±0.000	0.187±0.001	0.228±0.000	**0.189±0.000**	**0.222±0.000**	**0.231±0.001**	**0.223±0.000**	0.163±0.001	0.208±0.002	**0.251±0.002**	**0.339±0.001**	6
	MAE	0.247±0.000	0.260±0.001	0.278±0.000	0.313±0.001	0.259±0.001	0.283±0.000	0.292±0.001	0.285±0.000	0.209±0.000	0.250±0.001	0.287±0.000	0.341±0.000	0
PatchTST	MSE	0.180±0.002	0.187±0.000	0.204±0.000	0.246±0.001	0.227±0.001	0.258±0.000	0.275±0.000	0.276±0.001	0.172±0.001	0.220±0.002	0.280±0.000	0.355±0.000	0
	MAE	0.273±0.003	0.280±0.000	0.296±0.001	0.328±0.000	0.282±0.000	0.301±0.001	0.312±0.005	0.309±0.000	0.214±0.003	0.256±0.000	0.298±0.001	0.347±0.000	0
FEDFormer	MSE	0.220±0.001	0.237±0.001	0.259±0.000	0.310±0.001	0.278±0.002	0.345±0.001	0.372±0.001	0.344±0.002	0.220±0.001	0.278±0.001	0.340±0.001	0.401±0.000	0
	MAE	0.334±0.001	0.347±0.002	0.365±0.000	0.398±0.000	0.364±0.003	0.417±0.001	0.449±0.002	0.428±0.000	0.305±0.002	0.349±0.000	0.382±0.001	0.411±0.000	0
Autoformer	MSE	0.215±0.002	0.247±0.001	0.273±0.001	0.304±0.000	0.512±0.001	0.648±0.000	0.835±0.002	0.798±0.001	0.252±0.000	0.294±0.001	0.358±0.005	0.451±0.001	0
	MAE	0.324±0.004	0.347±0.000	0.371±0.001	0.396±0.000	0.505±0.001	0.609±0.002	0.669±0.001	0.693±0.000	0.325±0.003	0.353±0.000	0.391±0.001	0.459±0.000	0
TimesNet	MSE	0.191±0.001	0.205±0.001	0.258±0.003	0.295±0.007	0.236±0.002	0.271±0.000	0.337±0.001	0.315±0.000	0.169±0.002	0.225±0.003	0.282±0.000	0.360±0.002	0
	MAE	0.294±0.001	0.306±0.000	0.343±0.001	0.368±0.001	0.274±0.000	0.306±0.001	0.339±0.000	0.323±0.002	0.219±0.005	0.264±0.001	0.304±0.000	0.355±0.002	0
MICN	MSE	0.221±0.002	0.224±0.000	0.233±0.004	0.252±0.001	0.200±0.001	0.263±0.000	0.242±0.001	0.248±0.000	0.193±0.000	0.240±0.003	0.282±0.001	0.350±0.007	0
	MAE	0.322±0.003	0.328±0.000	0.337±0.001	0.354±0.000	0.273±0.001	0.305±0.001	0.290±0.000	0.289±0.000	0.250±0.003	0.301±0.000	0.331±0.002	0.387±0.001	0
TiDE	MSE	0.215±0.000	0.247±0.000	0.273±0.001	0.304±0.000	0.346±0.001	0.383±0.000	0.418±0.003	0.414±0.001	0.201±0.000	0.245±0.007	0.297±0.002	0.369±0.001	0
	MAE	0.324±0.001	0.347±0.003	0.371±0.000	0.396±0.000	0.362±0.002	0.379±0.000	0.390±0.001	0.380±0.000	0.240±0.008	0.275±0.002	0.311±0.001	0.357±0.0010	0
DLinear	MSE	0.210±0.000	0.210±0.000	0.223±0.001	0.258±0.000	0.290±0.002	0.320±0.000	0.352±0.001	0.356±0.003	0.196±0.001	0.239±0.002	0.281±0.000	0.345±0.002	0
	MAE	0.302±0.001	0.305±0.000	0.319±0.001	0.350±0.001	0.377±0.003	0.398±0.001	0.415±0.000	0.413±0.002	0.256±0.003	0.299±0.003	0.331±0.000	0.382±0.002	0

Table 6
Multivariate short-term time series forecasting results. The best results are highlighted in bold and the second best are underlined.

Models	Metric	METR -LA			PEMS -BAY		
		3	6	12	3	6	12
Ours	MAE	**2.55 ± 0.00**	**2.84 ± 0.01**	**3.29 ± 0.00**	**1.19 ± 0.00**	**1.51 ± 0.01**	**1.78 ± 0.00**
	RMSE	**5.06 ± 0.00**	**6.04 ± 0.01**	**7.10 ± 0.00**	**2.59 ± 0.00**	**3.62 ± 0.00**	**4.31 ± 0.01**
	MAPE(%)	**6.61 ± 0.02**	**8.11 ± 0.03**	**9.43 ± 0.01**	**2.61 ± 0.01**	**3.53 ± 0.03**	**4.36 ± 0.01**
ESG	MAE	2.68 ± 0.00	3.06 ± 0.01	3.49 ± 0.01	1.31 ± 0.02	1.63 ± 0.00	1.92 ± 0.00
	RMSE	5.15 ± 0.00	6.19 ± 0.02	7.23 ± 0.00	2.74 ± 0.00	3.71 ± 0.01	4.42 ± 0.01
	MAPE(%)	6.93 ± 0.04	8.20 ± 0.01	9.96 ± 0.02	2.76 ± 0.02	3.69 ± 0.02	4.52 ± 0.01
GWaveNet	MAE	2.69 ± 0.01	3.07 ± 0.00	3.53 ± 0.00	1.30 ± 0.00	1.63 ± 0.01	1.95 ± 0.01
	RMSE	5.15 ± 0.00	6.22 ± 0.01	7.37 ± 0.00	2.74 ± 0.01	3.70 ± 0.00	4.52 ± 0.01
	MAPE(%)	6.90 ± 0.02	8.37 ± 0.01	10.01 ± 0.02	2.73 ± 0.01	3.67 ± 0.00	4.63 ± 0.01
AGCRN	MAE	2.83 ± 0.00	3.20 ± 0.00	3.58 ± 0.00	1.35 ± 0.01	1.69 ± 0.00	1.96 ± 0.00
	RMSE	5.45 ± 0.01	6.55 ± 0.03	7.41 ± 0.01	2.83 ± 0.00	3.81 ± 0.01	4.52 ± 0.00
	MAPE(%)	7.56 ± 0.06	8.79 ± 0.01	10.13 ± 0.03	2.87 ± 0.03	3.84 ± 0.00	4.67 ± 0.01
MTGNN	MAE	2.69 ± 0.01	3.05 ± 0.00	3.49 ± 0.01	1.32 ± 0.01	1.65 ± 0.01	1.94 ± 0.00
	RMSE	5.18 ± 0.00	6.17 ± 0.00	7.23 ± 0.02	2.79 ± 0.01	3.74 ± 0.01	4.49 ± 0.00
	MAPE(%)	6.86 ± 0.02	8.19 ± 0.06	9.87 ± 0.01	2.77 ± 0.01	3.69 ± 0.04	4.53 ± 0.00
DCRNN	MAE	2.77 ± 0.00	3.15 ± 0.01	3.60 ± 0.00	1.38 ± 0.00	1.74 ± 0.01	2.07 ± 0.00
	RMSE	5.38 ± 0.00	6.45 ± 0.00	7.60 ± 0.01	2.95 ± 0.02	3.97 ± 0.00	4.74 ± 0.01
	MAPE(%)	7.30 ± 0.03	8.80 ± 0.05	10.50 ± 0.01	2.90 ± 0.01	3.90 ± 0.01	4.90 ± 0.00
Crossformer	MAE	2.69 ± 0.01	3.05 ± 0.00	3.48 ± 0.01	1.31 ± 0.00	1.63 ± 0.00	1.93 ± 0.01
	RMSE	5.17 ± 0.02	6.18 ± 0.01	7.27 ± 0.01	2.75 ± 0.00	3.69 ± 0.01	4.45 ± 0.00
	MAPE(%)	6.88 ± 0.05	8.21 ± 0.01	9.86 ± 0.03	2.72 ± 0.03	3.66 ± 0.07	4.49 ± 0.01

- **TCN-based methods:** MICN [39] utilizes TCN to extract both local and global features from time series data. TimesNet [9] utilizes Fourier Transform and two-dimensional convolution to learn temporal correlations across different periods.

Methods for modeling interdependencies. We selected 6 well-established forecasting models as benchmarks, including:

- **Methods without graphs.** Crossformer [11] is one of the representatives of Transformer-based models, employing cross-dimensional attention to model inter-series dependencies instead of learning graph structures.
- **Methods with a pre-defined graph.** DCRNN [15] proposes diffusion convolutional recurrent neural networks, which combine diffusion graph convolutions with recurrent neural networks.
- **Methods with adaptive learning graphs.** Graph WaveNet [40] proposes a spatial–temporal graph convolutional network, which integrates diffusion graph convolutions with 1D dilated convolution. AGCRN [2] proposes an adaptive recurrent graph convolution network. MTGNN [17] learns a graph-based convolutional model that employs an adaptive adjacency matrix to capture the correlation among time series.
- **Methods with dynamic graphs.** ESG [19] segments historical time series into a series of subsequences, independently learns the graph structure of each subsequence, and then employs RNNs for prediction.

5.1.4. Evaluation metrics

In these datasets, we use mean absolute error (MAE, defined in Eq. (17)) and Mean Squared Error (MSE, defined in Eq. (18)) to evaluate the accuracy of MTS forecasting.

$$\mathrm{MAE} = \frac{1}{nH}\sum_{i=1}^{n}\sum_{h=1}^{H}\left|y_{i,t+h} - \hat{y}_{i,t+h}\right|, \tag{17}$$

$$\mathrm{MSE} = \frac{1}{nH}\sum_{i=1}^{n}\sum_{h=1}^{H}\left(y_{i,t+h} - \hat{y}_{i,t+h}\right)^2, \tag{18}$$

where $y_{i,t+h}$ represents the actual value of the ith variable at time $t+h$, $\hat{y}_{i,t+h}$ represents the predicted value of the ith variable at time $t+h$, n is the number of variables in the time series, and H is the forecast horizon (number of time steps predicted).

5.2. Main results

Tables 3, 4, and 5 show the long-term forecasting results under two evaluation metrics: MAE and MSE, while Table 6 presents the short-term forecasting results using three metrics: MAE, RMSE, and MAPE.

As demonstrated in Table 3 and 5, PDFNet consistently surpasses PathFormer on datasets characterized by strong inter-variable correlations, such as Traffic, underscoring its distinctive capability to explicitly capture evolving interdependencies across multiple periods. Meanwhile, results presented in Table 4 indicate that PDFNet also delivers substantial performance improvements over PathFormer on the ETT datasets; however, these improvements are not uniformly observed across all evaluation metrics due to the relatively weaker inter-variable correlations inherent in these datasets. This outcome aligns with the inherent design of PDFNet, as explicit modeling of dynamic interactions naturally yields greater benefits when pronounced inter-variable dependencies exist. Additionally, as evidenced by the results in Table 3, PathFormer achieves superior forecasting accuracy on the ILI dataset, primarily because this dataset is notably small and exhibits minimal inter-variable dependencies, thus constraining the advantages gained by PDFNet's correlation-aware modeling. Taken together, these empirical findings validate the underlying assumptions of PDFNet and highlight its particular effectiveness in scenarios with significant variable interactions.

PatchTST and MICN have demonstrated excellent performance on certain datasets, as both models effectively learn local temporal features akin to periodic behavior. Similarly, FEDformer and TimesNet have shown strong performance on various datasets by leveraging frequency domain information to capture global, long-term temporal features. These results highlight the significance of incorporating both local and global temporal information in forecasting. However, its inability to utilize dependencies among variables leads to suboptimal performance in datasets characterized by a large number of variables and strong interdependencies. PDFNet excels in integrating these aspects by capturing features both within and across periods.

As illustrated in Table 6, PDFNet achieves state-of-the-art performance compared to models that primarily focus on inter-series dependencies. This is because our model effectively learns changes within different periods as well as the evolution between periods, utilizing key periodic features in multivariate time series (MTS) and capturing the dependencies among variables. We observed that models such as Graph WaveNet, AGCRN, and MTGNN, which adaptively learn static graphs, outperform DCRNN, which relies on manually constructed prior graphs. This finding underscores the effectiveness of data-driven approaches in learning graph structures. Crossformer surpasses most models by effectively utilizing the Transformer architecture to learn interdependencies among time series. This performance highlights the critical importance of learning these dependencies for improved predictive accuracy. ESG achieved suboptimal performance, which indicates that learning dynamic graphs is advantageous for capturing the interdependencies associated with dynamic changes. These methods do not capture the periodic features that are crucial for time series forecasting. In contrast, PDFNet effectively learns intra-period variations and inter-period interdependencies, thereby outperforming the baseline methods.

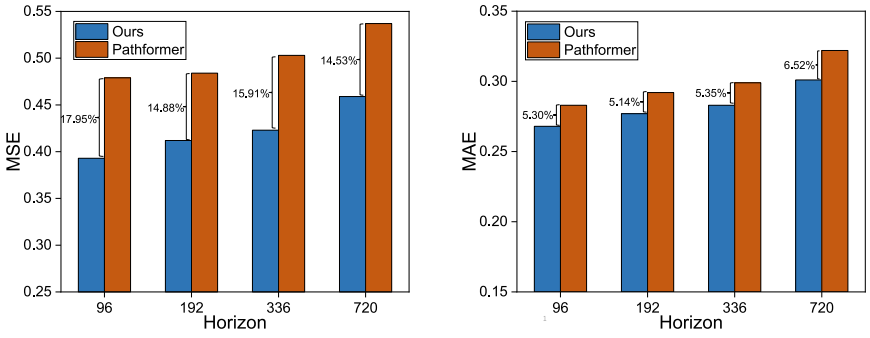
Figs. 4 and 5 presents a comparison of PDFNet and PathFormer, on the Traffic and Electricity datasets for two metrics. For the Traffic dataset, PDFNet achieved an average reduction of 15.82% and 5.58% in MSE and MAE, respectively. For the Electricity dataset, PDFNet achieved an average reduction of 7.05% and 1.5% in MSE and MAE, respectively. The experimental results demonstrate that PDFNet improves performance across various forecasting horizons, proving the effectiveness of our method in enhancing predictive performance.

5.3. Ablation studies

We conducted ablation studies to verify the effectiveness of the proposed modules. We compared PDFNet with the following variants:

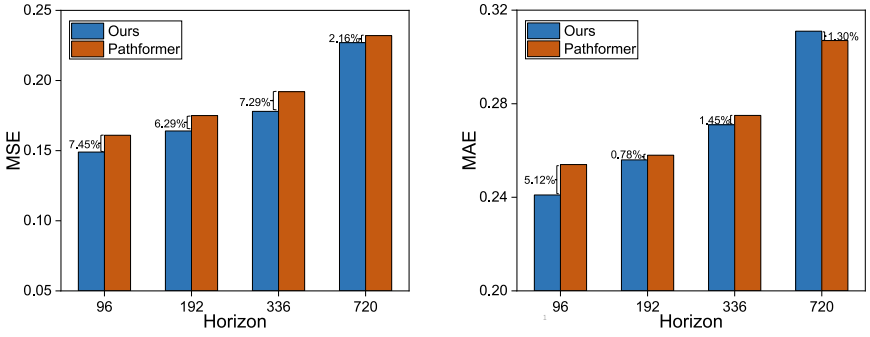
- w/o GatedDG. This variant does not utilize our proposed gated dynamic graph structure learning module; instead, it directly outputs the results from the self-attention module.
- w/o Intra-period. In this variant, the modeling of intra-period variations is removed, isolating the effect of inter-period interactions.
- w/o Inter-period: This variant excludes the inter-period interaction component from the gated dynamic graph module, isolating the effect of intra-period interactions.
- w/o Self-attention. This variant does not employ the self-attention mechanism; instead, it directly feeds the results from the Conv1D into the Gated DGSL.
- w/o Conv1D. This variant does not apply Conv1D to extract local features from each patch; instead, it directly inputs the embedded results into the self-attention module.

Table 7 presents the forecasting results of different variants of PDFNet on the Electricity, Solar Energy, and Exchange Rate datasets. From the table, it is evident that the variant without the GatedDG performs the worst, which demonstrates that learning intra-period variations and inter-period interactions enhances the predictive performance of the model. Particularly in multivariate datasets with strong



(a) MSE (b) MAE

Fig. 4. Comparison results of MSE, MAE, and RMSE between PDFNet and PathFormer on the Traffic dataset.



(a) MSE (b) MAE

Fig. 5. Comparison results of MSE, MAE, and RMSE between PDFNet and PathFormer on the Electricity dataset.

Table 7
Ablation studies.

Dataset	Electricity			Solar Energy			Exchange		
Method	MSE	MAE	RMSE	MSE	MAE	RMSE	MSE	MAE	RMSE
PDFNet	**0.167**	**0.260**	**0.409**	**0.316**	**0.336**	**0.562**	**0.177**	**0.300**	**0.421**
w/o GatedDG	0.267	0.351	0.517	0.567	0.514	0.753	0.223	0.341	0.473
w/o Intra-Period	0.212	0.312	0.457	0.362	0.367	0.602	0.207	0.308	0.436
w/o Inter-Period	0.219	0.317	0.463	0.379	0.372	0.617	0.213	0.312	0.442
w/o Self-attention	0.189	0.271	0.434	0.346	0.343	0.589	0.200	0.320	0.447
w/o Conv1D	0.182	0.271	0.427	0.323	0.333	0.568	0.191	0.316	0.437

spatiotemporal correlations, the improvement is more pronounced. For instance, the Exchange dataset, with only eight variables, shows a relatively smaller enhancement compared to the other two datasets. Notably, the ablation results further reveal that removing either the intra-period or the inter-period component leads to a significant degradation in forecasting accuracy, underscoring the complementary roles of these two interaction types in capturing the complex dependencies inherent in multivariate time series data. The prediction results of the variant without self-attention are consistently better across all datasets compared to the variant without Conv1D, highlighting the effectiveness of the attention mechanism in capturing long-term temporal features. PDFNet achieves the best performance across all datasets, further demonstrating that considering both local and global information, as well as intra-period variations and inter-period interactions, is crucial for multivariate time series forecasting.

5.4. Hyperparameter sensitivity analysis

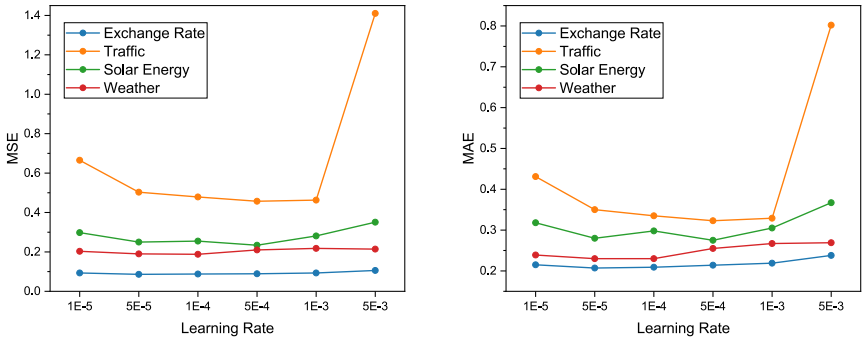
To thoroughly assess the stability of PDFNet, we conducted sensitivity analyses on two key sets of hyperparameters: one related to the optimization process and the other intrinsic to the model structure. In the following subsections, we present our findings on learning rate sensitivity and the impact of the selected frequency components on forecasting performance.

5.4.1. Learning rate sensitivity

We first analyzed the influence of the learning rate on model performance. The results indicate that both smaller and larger learning rates lead to performance degradation, necessitating the careful selection of an appropriate value. Specifically, on the Traffic dataset, an increase in the learning rate to $5E-3$ resulted in a sharp rise in the values of MSE and MAE. The model achieves stable and optimal performance within a moderate learning rate range. Detailed performance metrics (MAE and MSE) across different learning rates are provided in Fig. 6.

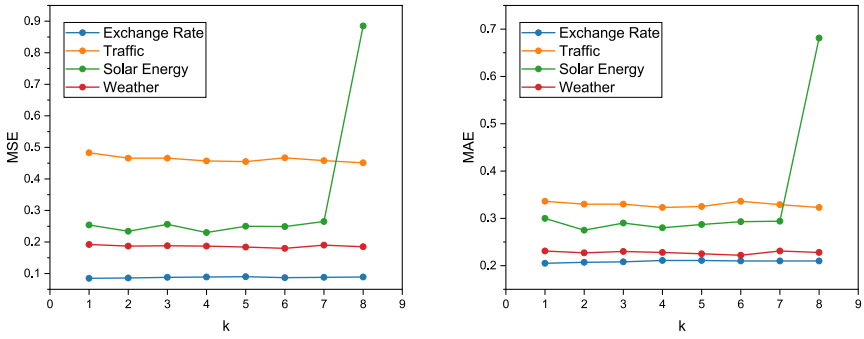
5.4.2. Sensitivity to selected frequency components

We evaluated the effect of the number of selected frequency components—corresponding to the number of periods extracted via



(a) MSE (b) MAE

Fig. 6. The impact of different learning rates on forecasting performance.



(a) MSE (b) MAE

Fig. 7. The impact of selected frequency components on predictive performance.

frequency-domain analysis—on model performance. In our initial analysis, the sensitivity of this hyperparameter was only briefly addressed. Here, we extend this evaluation by systematically varying the number of selected frequency components from a minimal to a higher count. Our experiments reveal that an optimal range exists where the model effectively captures the dominant periodic features, whereas selecting too few or too many frequency components leads to an insufficient or noisy representation of the underlying periodic patterns. The impact on forecasting accuracy is quantified in Fig. 7, demonstrating that an appropriate selection of frequency components is crucial for maximizing the benefits of our dynamic graph modeling.

5.5. Visualization

5.5.1. Visualization of prediction results

We present a comprehensive evaluation of eight models across datasets from four distinct domains: Energy, Economy, Electricity, and Transportation. The models, which include PDFNet, PatchTST, Autoformer, FEDformer, TimesNet, MICN, DLinear, and TiDE, are analyzed for their performance across various forecast horizons, as illustrated in Figs. 8 to 9. Our results highlight the superior performance and accuracy of PDFNet across all domains and forecast lengths. Notably, PDFNet excels in the Transportation domain, capturing the time series trends more accurately than its counterparts. In contrast, other models struggled with this dataset, which exhibits strong variable interdependencies.

In the Weather dataset, for long-range forecasts up to 720 time steps, PDFNet achieved the most stable outcomes. Autoformer significantly deviated from the actual values, while FEDformer and MICN exhibited considerable fluctuations. TimesNet failed to capture the time series trends, likely due to its inability to learn periodicities beyond the look-back window. While PatchTST, DLinear, and TiDE identified trends within the series, they fell short in capturing the finer details within short periodic fluctuations.

5.5.2. Visualization of dynamic periodic changes

As depicted in Fig. 10, our study provides heatmap visualizations across distinct periods within the traffic dataset. Each heatmap captures the correlations between variables specific to its respective period. As further illustrated by the line plots in Fig. 10, the correlation strength between node pairs (6,4) and (5,20) dynamically fluctuates across periods, exhibiting notable periods of intensification and attenuation. In contrast, node pair (1,2) maintains consistently strong correlations throughout all periods, whereas node pair (10,20) consistently demonstrates weak correlations. This observation suggests that our model is capable of capturing both the changing and invariant aspects across different periods.

5.6. Comparison of inference time

As shown in Table 8, we recorded the inference time for different models on test datasets. The inference time of PDFNet is substantially lower than that of PathFormer. This significant advantage arises because PathFormer requires considerable computational overhead to search for optimal temporal scales through extensive parameter tuning, whereas our proposed PDFNet directly mines inherent multi-period features from the multivariate time series data. Among all evaluated methods, TimesNet consistently exhibits relatively long inference duration across datasets, primarily due to its heavy reliance on two-dimensional convolutional layers, resulting in an exceptionally large number of parameters. In contrast, MLP-based models such as TimeMixer and DLinear achieve the shortest inference times, benefiting from their simple linear network structures that significantly reduce computational complexity.

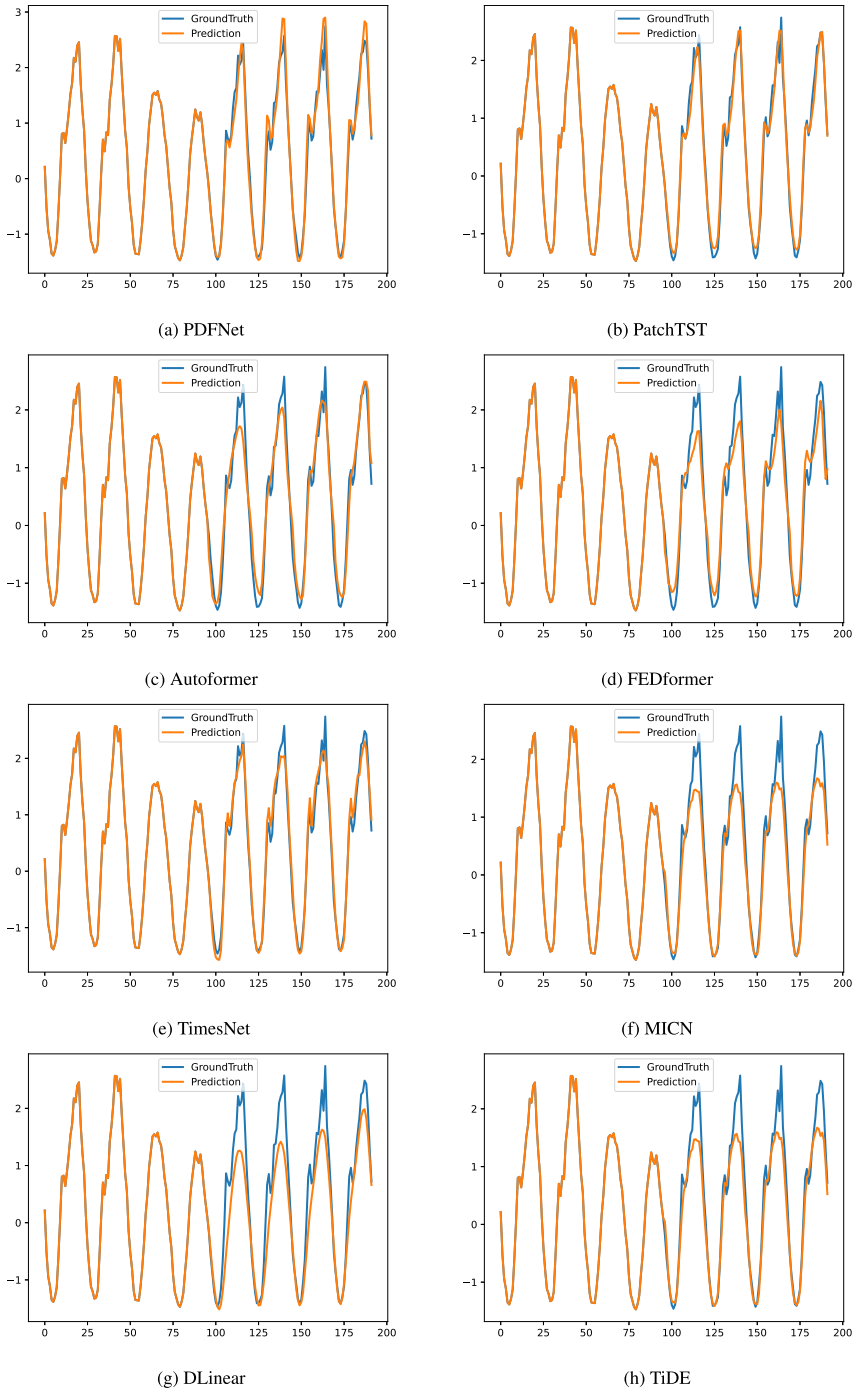


Fig. 8. Visualization of the prediction results was conducted using both a look-back window and a forecast horizon of 96 on the Traffic dataset.

6. Conclusion

In this paper, we focus on modeling the interdependencies among variables that evolve with periodic cycles, demonstrating unique challenges that are overlooked by existing mainstream time series models. Therefore, we introduce the Periodic Dynamics Forecasting Network (PDFNet), designed to tackle the challenges of modeling multi-period dynamic interdependencies in multivariate time series. Our proposed method effectively leverages the frequency and amplitude characteristics of time series data to adaptively identify and capture their underlying multi-periodic patterns. Furthermore, we develop a novel combination of a gated periodic recurrent unit and a gated graph structure learning module facilitating the construction of dynamic graphs that encapsulate the complex interdependencies among variables.

Extensive experiments conducted on multiple MTS datasets validate the superior performance achieved by PDFNet and highlight the effectiveness of capturing intra-period variations and inter-period dependencies among multivariate time series. These results underscore the importance of accounting for interdependencies among variables across different periods and modeling the interactions between these periods.

As a limitation, our approach assumes that the periodic durations are identical across different variables. Each time series in real-world scenarios exhibits unique periodic variations, including durations and

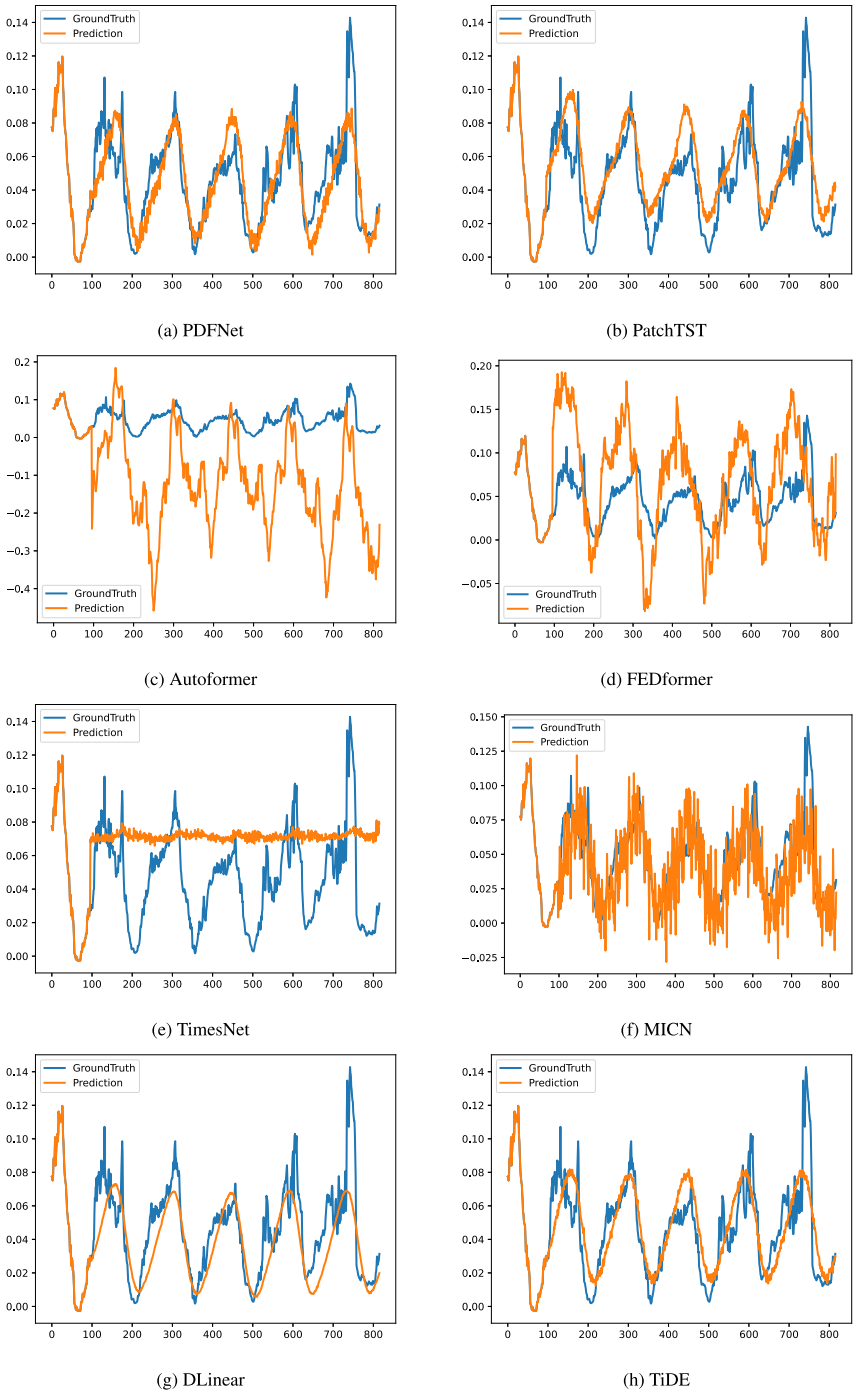


Fig. 9. Visualization of the prediction results was conducted using a look-back window of 96 and a forecast horizon of 720 on the Weather dataset.

amplitudes, and the periodic variations between variables are highly interrelated. Further exploration of varying durations of each period in different time series would be challenging yet intriguing. We hope that future research can address this challenge with reduced computation complexity based on the proposed architecture.

CRediT authorship contribution statement

Gang Tan: Writing – original draft, Methodology, Data curation, Conceptualization. **Yueyang Wang:** Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Ziyi Xiao:** Writing – review & editing, Validation, Data curation. **Dandan He:** Writing – review & editing, Visualization, Methodology. **Guodong Sa:** Visualization, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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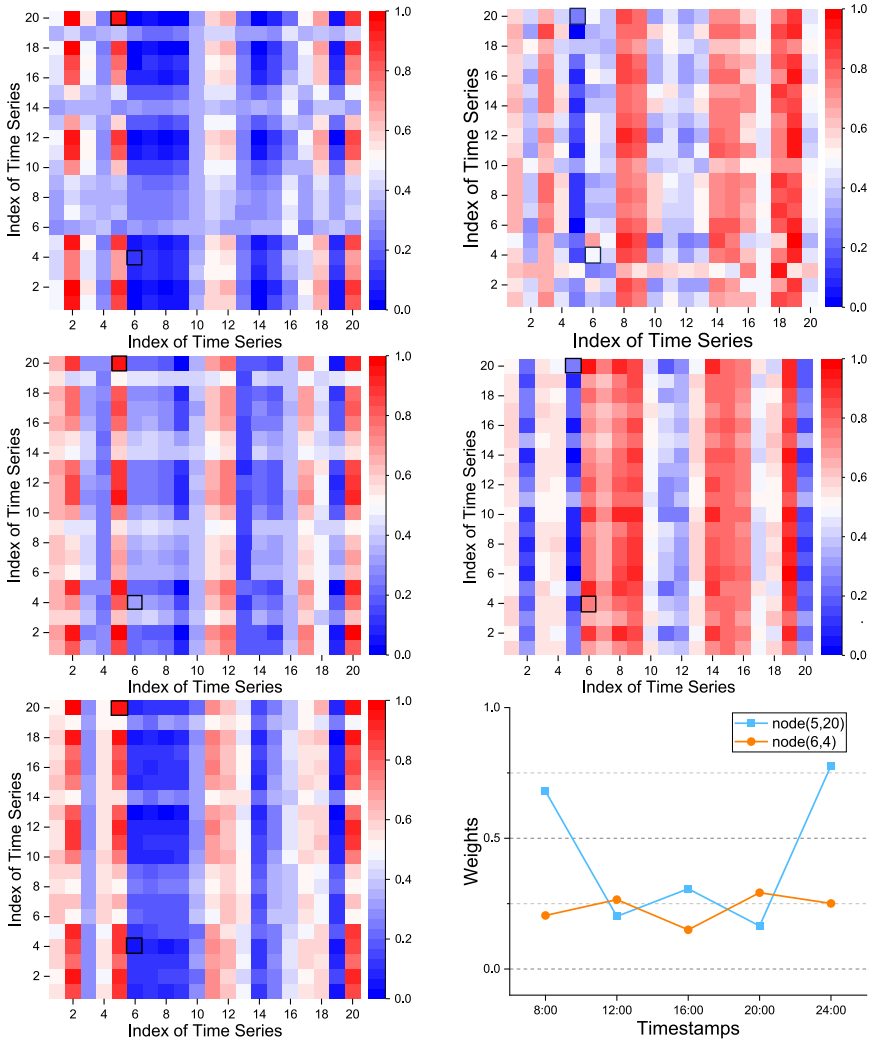


Fig. 10. Visualization of inter-variable correlation heatmaps across periods on the Traffic dataset.

Table 8
Results of average inference time consumption (in seconds).

Models		PDFNet	Pathformer	TimeMixer	PatchTST	TimesNet	MICN	DLinear
ETTm1	96	14.379	56.135	6.683	5.644	34.672	9.924	3.395
	192	15.088	56.561	6.917	6.299	37.963	4.408	3.868
	336	15.612	56.626	7.401	6.598	26.795	4.969	4.316
	720	16.481	54.521	8.236	6.502	73.803	5.969	5.906
	Avg	15.39	55.961	7.309	6.261	43.308	6.318	4.371
Exchange	96	10.516	64.396	25.870	19.949	145.556	8.334	11.041
	192	17.673	63.828	34.977	25.428	235.933	11.409	16.888
	336	29.377	44.596	56.400	47.102	311.407	18.787	37.407
	720	90.047	56.224	80.916	47.018	574.837	66.857	72.595
	Avg	36.903	57.261	49.541	34.874	316.933	26.347	34.483
ILI	24	0.608	1.293	0.705	0.700	0.705	0.584	0.503
	36	0.566	1.982	0.686	0.693	0.686	0.570	0.486
	48	0.598	1.064	0.650	0.707	0.650	0.615	0.514
	60	0.593	1.068	0.670	0.709	0.670	0.582	0.458
	Avg	0.591	0.352	0.678	0.702	0.678	0.588	0.490

Data availability

Data will be made available on request.

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